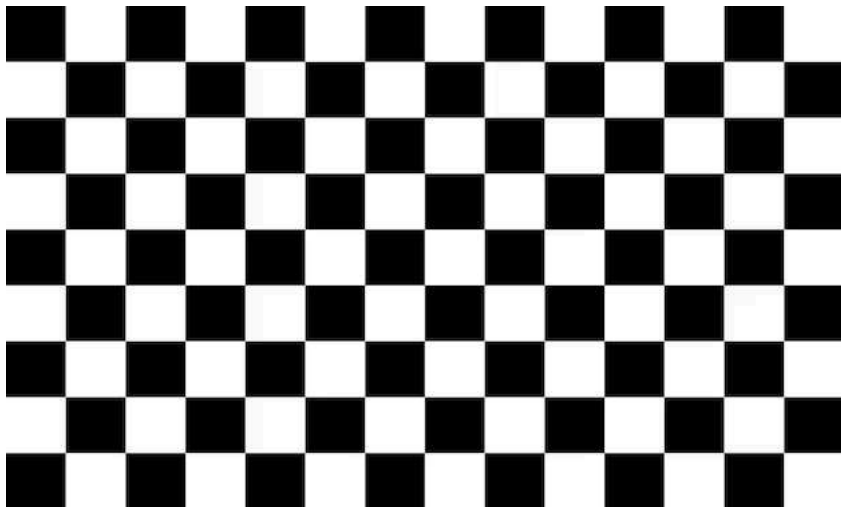


```
Image(filename="/content/checkerboard.png")
```

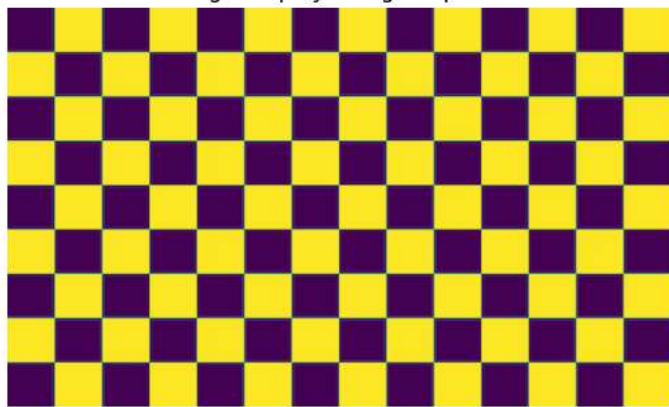


```
[ 0 0 0 ... 255 255 255]
[ 0 0 0 ... 255 255 255]
[ 0 0 0 ... 255 255 255]
...
[ 0 0 0 ... 255 255 255]
[ 0 0 0 ... 255 255 255]
[ 0 0 0 ... 255 255 255]
```

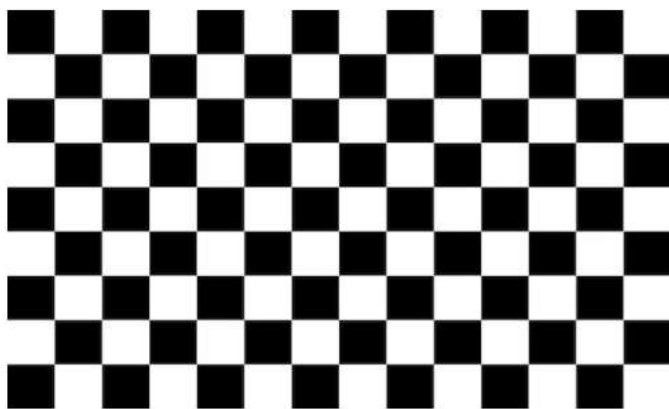
```
Shape: (376, 626)
Dtype: uint8
Size: 235376
```

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Image Display Using Matplotlib



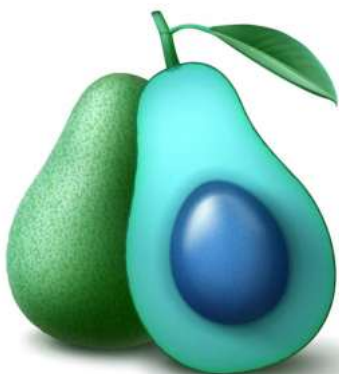
```
plt.imshow(cb_img, cmap = "gray")  
plt.axis("off")  
plt.show()
```



```
avacado_img = cv2.imread("avacado.jpg", cv2.IMREAD_COLOR)  
print("Image shape (height, width, channels) is:", avacado_img.shape)  
print("Image size is: ", avacado_img.size)  
print("Data type of image is:", avacado_img.dtype)
```

```
Image shape (height, width, channels) is: (740, 740, 3)  
Image size is: 1642800  
Data type of image is: uint8
```

```
plt.imshow(avacado_img)  
plt.axis("off")  
plt.show()
```



```
avacado_img_channels_reversed = avacado_img[:, :, ::-1]
plt.imshow(avacado_img_channels_reversed)
plt.axis("off")
plt.show()
```



```
b, g, r = cv2.split(avacado_img)
plt.figure(figsize = [20, 5])
plt.subplot(1, 4, 1)
plt.imshow(r, cmap = "gray")
plt.title("Red Channel")
plt.axis("off")
plt.subplot(1, 4, 2)
plt.imshow(g, cmap = "gray")
plt.title("Green Channel")
plt.axis("off")
plt.subplot(1, 4, 3)
plt.imshow(b, cmap = "gray")
plt.title("Blue Channel")
plt.axis("off")
merged_img = cv2.merge((b, g, r))
plt.subplot(1, 4, 4)
plt.imshow(merged_img[:, :, ::-1])
plt.title("Merged Output")
plt.axis("off")
plt.show()
```

Red Channel



Green Channel



Blue Channel



Merged Output



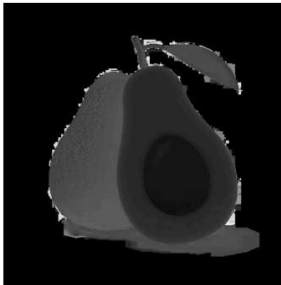
```
avacado_img_bgr = cv2.imread("avacado.jpg", cv2.IMREAD_COLOR)
avacado_img_rgb = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2RGB)
plt.imshow(avacado_img_rgb)
plt.title("BGR to RGB")
plt.axis("off")
plt.show()
```

BGR to RGB

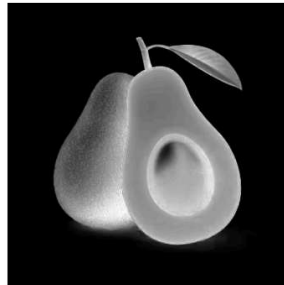


```
avacado_img_hsv = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2HSV)
h, s, v = cv2.split(avacado_img_hsv)
plt.figure(figsize = [20, 5])
plt.subplot(1, 4, 1)
plt.imshow(h, cmap = "gray")
plt.title("H Channel")
plt.axis("off")
plt.subplot(1, 4, 2)
plt.imshow(s, cmap = "gray")
plt.title("S Channel")
plt.axis("off")
plt.subplot(1, 4, 3)
plt.imshow(v, cmap = "gray")
plt.title("V Channel")
plt.axis("off")
plt.subplot(1, 4, 4)
plt.imshow(avacado_img_rgb)
plt.title("Original")
plt.axis("off")
plt.show()
```

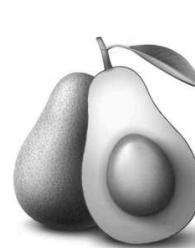
H Channel



S Channel



V Channel

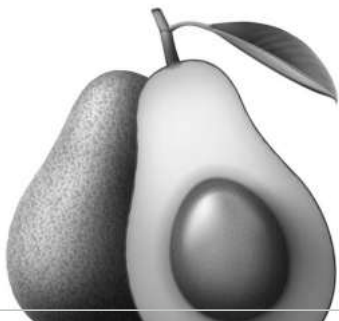


Original



```
avacado_img_gray = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2GRAY)
plt.imshow(avacado_img_gray, cmap = "gray")
plt.title("BGR to grayscale")
plt.axis("off")
plt.show()
```

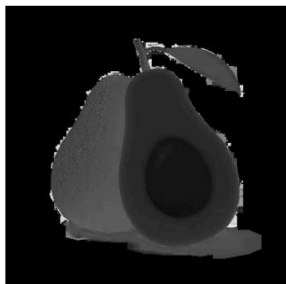
BGR to grayscale



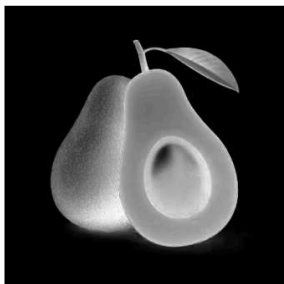
```
h_new = h + 30
avacado_img_merged = cv2.merge((h_new, s, v))
avacado_img_rgb = cv2.cvtColor(avacado_img_merged, cv2.COLOR_HSV2RGB)
h,s,v = cv2.split(avacado_img_merged)
plt.figure(figsize = [20, 5])
plt.subplot(1, 4, 1)
plt.imshow(h, cmap = "gray")
plt.title("H + 30 Channel")
plt.axis("off")
plt.subplot(1, 4, 2)
plt.imshow(s, cmap = "gray")
plt.title("S Channel")
plt.axis("off")
plt.subplot(1, 4, 3)
plt.imshow(v, cmap = "gray")
plt.title("V Channel")
plt.axis("off")
plt.subplot(1, 4, 4)
plt.imshow(avacado_img_rgb)
plt.title("Merged Image")

plt.axis("off")
plt.show()
```

H + 30 Channel



S Channel



V Channel



Merged Image



```
cv2.imwrite("avacado_saved.jpg", avacado_img_rgb)
```

```
True
```

Start coding or [generate](#) with AI.